Chapter 1: Objectives

1. Primary Objective

To develop an intelligent, evidence-based, and personalized clinical nutrition and fitness assistant that delivers tailored meal plans, caloric targets, and workout guidance while strictly enforcing medical safety, allergen avoidance, and user health constraints.

2. Specific Functional & Technical Objectives

A. Personalized Health & Nutritional Profiling

- **Automate Biometric Assessments**: Calculate accurate physiological metrics—such as Body Mass Index (BMI), Basal Metabolic Rate (BMR using the Mifflin-St Jeor equation), and Total Daily Energy Expenditure (TDEE)—based on user age, gender, height, weight, and daily activity level.
- **Tailored Macro & Caloric Planning**: Dynamically generate personalized daily caloric limits and macronutrient distributions (protein, carbohydrates, healthy fats) aligned with the user's primary fitness goal (e.g., sustainable fat loss, lean muscle gain, weight maintenance).
- **Incorporate Lifestyle & Cultural Preferences**: Factor in individual dietary classifications (Vegetarian, Vegan, Non-Vegetarian, Eggetarian) alongside cultural cuisine choices and taste preferences.

B. Evidence-Based AI with Clinical Safety Guardrails

- **Ground AI Responses via RAG (Retrieval-Augmented Generation)**: Integrate vector retrieval (ChromaDB + HuggingFace embeddings) with an LLM (via Groq) to ground answers in verified nutritional data rather than unvetted generic web data, minimizing hallucinations.
- **Enforce Zero-Tolerance Allergen Filters**: Programmatically and contextually detect and eliminate all user-specified allergens and their derivatives from all recommended foods and meal plans.
- **Medical Condition Adaptation**: Adapt dietary recommendations to accommodate chronic health conditions such as Type 2 Diabetes, PCOS, Hypertension, Thyroid disorders, and GERD.
- **Prevent Harmful Health Advice**: Prohibit extreme, unscientific crash diets, avoid diagnostic claims or drug prescriptions, and enforce transparent medical disclaimers directing users to certified medical professionals for clinical diagnoses.

C. Smooth, Context-Aware User Experience

- **Maintain Multi-Turn Conversational Memory**: Implement a rolling context window (e.g., retrieving the last 6 messages from MongoDB) so the AI can handle follow-ups, substitutions, and iterative plan refinements without losing track of previous statements.
- **Structured, Readable Information Delivery**: Deliver complex multi-day diet and workout routines using clean, responsive Markdown tables with ingredient quantities, preparation swaps, sets, reps, and rest intervals.

D. Scalable, Secure Architecture

- **Dual-Database Optimization**:
    - Use a relational database (PostgreSQL) for storage of user profiles, authentication credentials, and login audit histories.
    - Use a document database (MongoDB) for flexible, high-volume storage of chat sessions and conversation message streams.
- **Data Protection & Access Control**: Secure user credentials using modern cryptographic password hashing (Bcrypt), session verification decorators, and timestamped audit logs.

Chapter 2: Scope & Applicability

1. Project Scope

A. User Biometrics & Mathematical Profiling:

- Captures primary physical parameters: age, gender, height, weight, activity tiers, and fitness goals.
- Computes standard physiological indexes: BMI, BMR (via Mifflin-St Jeor formula), TDEE, target daily calorie intake, and daily hydration estimates.

B. Context-Aware Conversational Interface:

- Natural language interaction powered by LLM inference (via Groq) and LangChain.
- Conversational memory holding a rolling buffer (last 6 exchanges) to enable natural follow-up questions, substitutions, and plan iterations.

C. Retrieval-Augmented Generation (RAG):

- Grounded knowledge retrieval using local ChromaDB vector store and HuggingFace semantic embeddings to query verified nutrition guidelines, reducing hallucinations.

D. Custom Dietary & Lifestyle Customization:

- Filters meal options based on dietary choices (Veg, Non-veg, Vegan, Eggetarian).
- This tailors suggestions around cultural cuisines (Indian, Mediterranean, etc.) and taste restrictions.

E. Preventive Safety & Allergen Guardrails:

- Zero-tolerance exclusions for specified allergens (dairy, peanuts, gluten, shellfish, etc.).
- General nutritional adjustments for common lifestyle conditions (PCOS, Type 2 Diabetes, Hypertension, Thyroid, Acid Reflux/GERD).
- Mandatory medical disclaimer prompts and acknowledgement workflows.

F. Multi-Database Management:

- Structured user profiles and encrypted auth credentials stored in PostgreSQL.
- Dynamic, multi-session chat histories and timestamped message logs stored in MongoDB.

2. Applicability

A. General Public & Fitness Enthusiasts

- **Sustainable Body Composition Goals**: Individuals looking to lose fat, build muscle, or maintain weight without resorting to extreme, unscientific crash diets.

- **Portion & Macro Guidance**: Busy professionals or students who need quick, structured grocery ideas and portion estimates (grams, cups) suited to their busy schedules.

B. Individuals with Dietary Restrictions & Allergies

- **Allergen-Safe Meal Planning**: Individuals who struggle to find recipes that strictly avoid allergens like gluten, dairy, nuts, or soy.

C. Wellness Clinics, Gyms & Fitness Centres

- **Trainer & Nutritionist Triage**: Gyms, wellness centres, and corporate wellness programs can deploy the platform as a baseline tier-1 assessment tool, freeing human coaches to focus on high-touch coaching.

Chapter 3: Technologies Used

1. Programming & Backend Framework

- Python 3.10.20
- Flask
- Jinja2

2. Artificial Intelligence & RAG Pipeline

- ChatGroq (Groq LPU™ Inference Engine with model openai/gpt-oss-20b)
- LangChain
- ChromaDB
- HuggingFace Embeddings

3. Database & Polyglot Storage Architecture

- Relational Database (RDBMS) — PostgreSQL:
  - **Engine**: PostgreSQL
  - **ORM**: Flask-SQLAlchemy (Object-Relational Mapping for structured relational models).
- Document Database (NoSQL) — MongoDB:
  - **Driver**: PyMongo (MongoClient)
  - **Database**: DietMate_DB

4. Security & Authentication

- Flask-Bcrypt
- Python functools.wraps
- Audit Logging

5. Frontend Technologies

- HTML5
- CSS3
- JavaScript & jQuery 3.5.1
- Bootstrap 4.5.2
- Google Fonts

6. Data Parsing & Utilities

- Markdown & MarkupSafe:
  - markdown (Python)
  - markupsafe.Markup
- Python dotenv (python-dotenv)
- Python threading & webbrowser

Chapter 4: Project Module

Module 1: User Authentication & Security Module

- **User Registration (/signup)**: Validates input uniqueness, email uniqueness, and encrypts passwords using Flask-Bcrypt with salted hashes.
- **User Login (/login)**: Verifies credentials, initializes secure user sessions (session['user_id']), and intelligently routes users (redirecting to /chat if a profile exists, or /profile for new users).
- **Session Protection (login_required Decorator)**: Python decorator wrapper safeguarding all private endpoints (/profile, /chat, /get_chat_history) against unauthenticated direct URL access.
- **User Logout (/logout)**: Records exit timestamps before terminating active session states.

Module 2: Biometric Assessment & Health Profiling Module

- **Client-Side Physiological Calculator**:
  - **BMI (Body Mass Index)**: Metric calculation with dynamic clinical classification (Underweight, Normal, Overweight, Obese).
  - **BMR (Basal Metabolic Rate)**: Computed using the clinically verified **Mifflin-St Jeor formula**.
  - **TDEE & Target Caloric Allowance**: Factors in user-selected physical activity multipliers (Sedentary to Very Active) and fitness goals (Fat Loss, Maintenance, Muscle Gain).
  - **Macronutrient & Hydration Split**: Calculates daily grams of Protein, Carbohydrates, Healthy Fats, and recommended daily water intake (liters).
- **Profile Customization Engine:**
  - **Dietary Preferences**: Categorizes dietary types (Vegetarian, Vegan, Non-Vegetarian, Eggetarian).
  - **Cultural Cuisines & Taste**: Captures preferences (Indian, Mediterranean, Low Carb, Halal, etc.).
  - **Clinical Guardrails**: Captures user-declared allergens (peanuts, dairy, gluten, shellfish, etc.) and chronic conditions (Diabetes, PCOS, Thyroid, Hypertension, GERD).
  - **Mandatory Medical Acknowledgement**: Checkbox validation ensuring users acknowledge that DietMate AI provides wellness advice, not clinical diagnoses.

Module 3: Clinical RAG (Retrieval-Augmented Generation) & AI Engine

- **Vector Database Indexing (ChromaDB)**: Local vector database housing indexed chunks of evidence-based dietary literature.
- **Semantic Embedding Pipeline**: Pre-trained HuggingFace sentence transformers translating queries and documents into high-dimensional vector representations.
- **Retrieval Pipeline (retriever)**: Executes similarity searches to pull the top k=3 most relevant knowledge chunks.
- **Inference Orchestration (Groq & LangChain)**:
- Leverages ChatGroq (openai/gpt-oss-20b) for deterministic, clinical precision.
- Uses create_stuff_documents_chain and create_retrieval_chain to combine retrieved context with user input.
- **System Prompt Guardrails (system_prompt)**:
  - Enforces zero-tolerance allergen exclusions.
  - Adapts meal plans for specific medical conditions.
  - Strictly prohibits extreme crash diets or unauthorized prescription advice.

Module 4: Conversational Memory & Multi-Turn Context Buffer

- **Rolling Context Buffer**: Queries MongoDB to retrieve the last 6 messages (3 user queries + 3 assistant responses) for the active chat_id.
- **Chronological Reconstruction**: Re-orders past messages sequentially from oldest to newest.

Module 5: Interactive Chat & Presentation Module

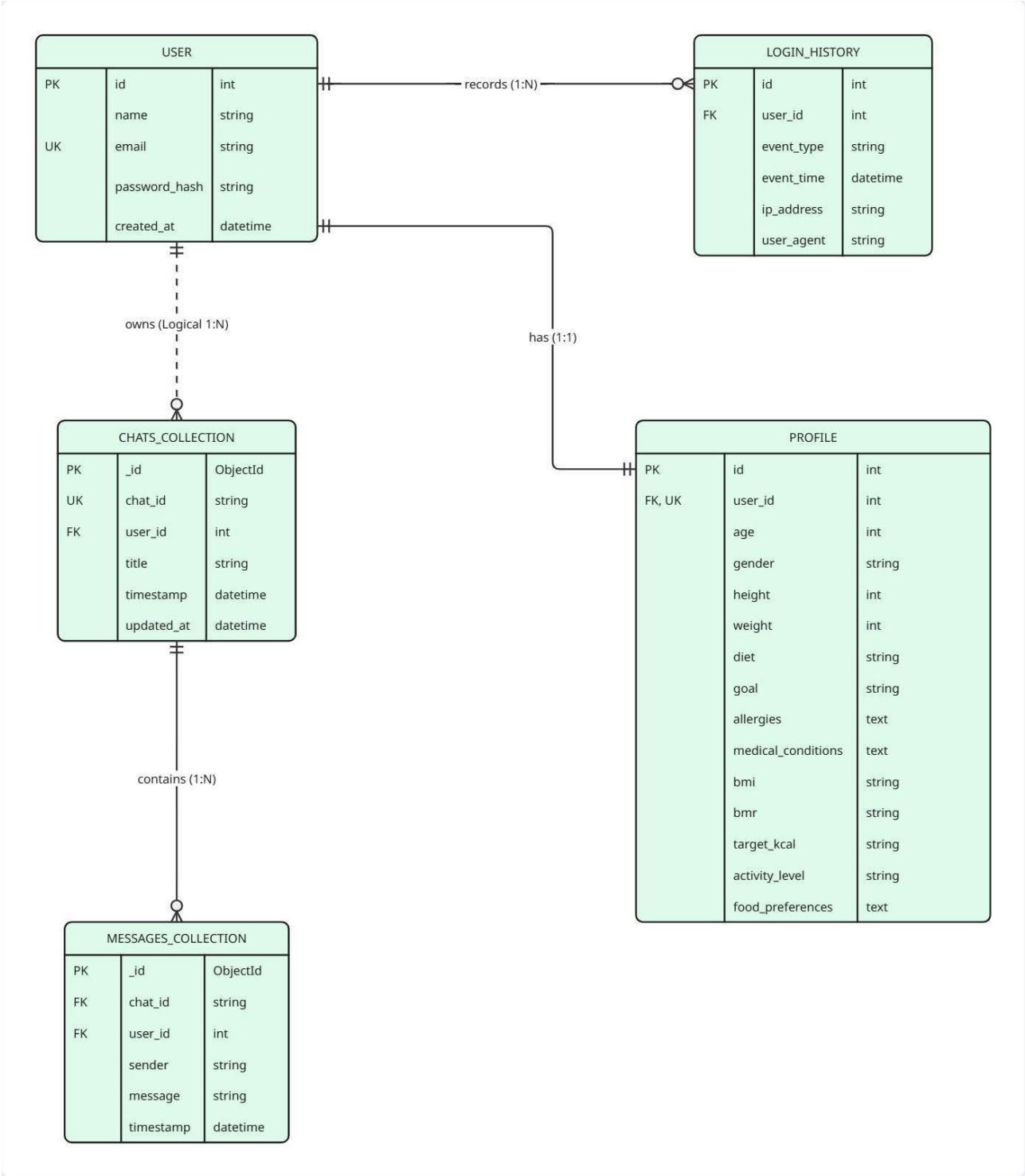
- **Dynamic Time-of-Day Greeting**: Automatically updates greeting headlines (*"Good morning"*, *"Good afternoon"*, *"Good evening"*, *"Hello"*) based on the hour.
- **Chat History Sidebar**: Displays past conversation sessions fetched from MongoDB, allowing users to switch threads or start a "New Chat".
- **Markdown & Table Rendering**: Employs Python markdown to format meal and workout schedules into clean tables.
- **Interactive Suggestion Chips**: One-click quick-start prompts for common wellness tasks (7-day diet plan, beginner workout, high-protein foods, sustainable fat loss).

Module 6: Polyglot Persistence & Audit Logging Module

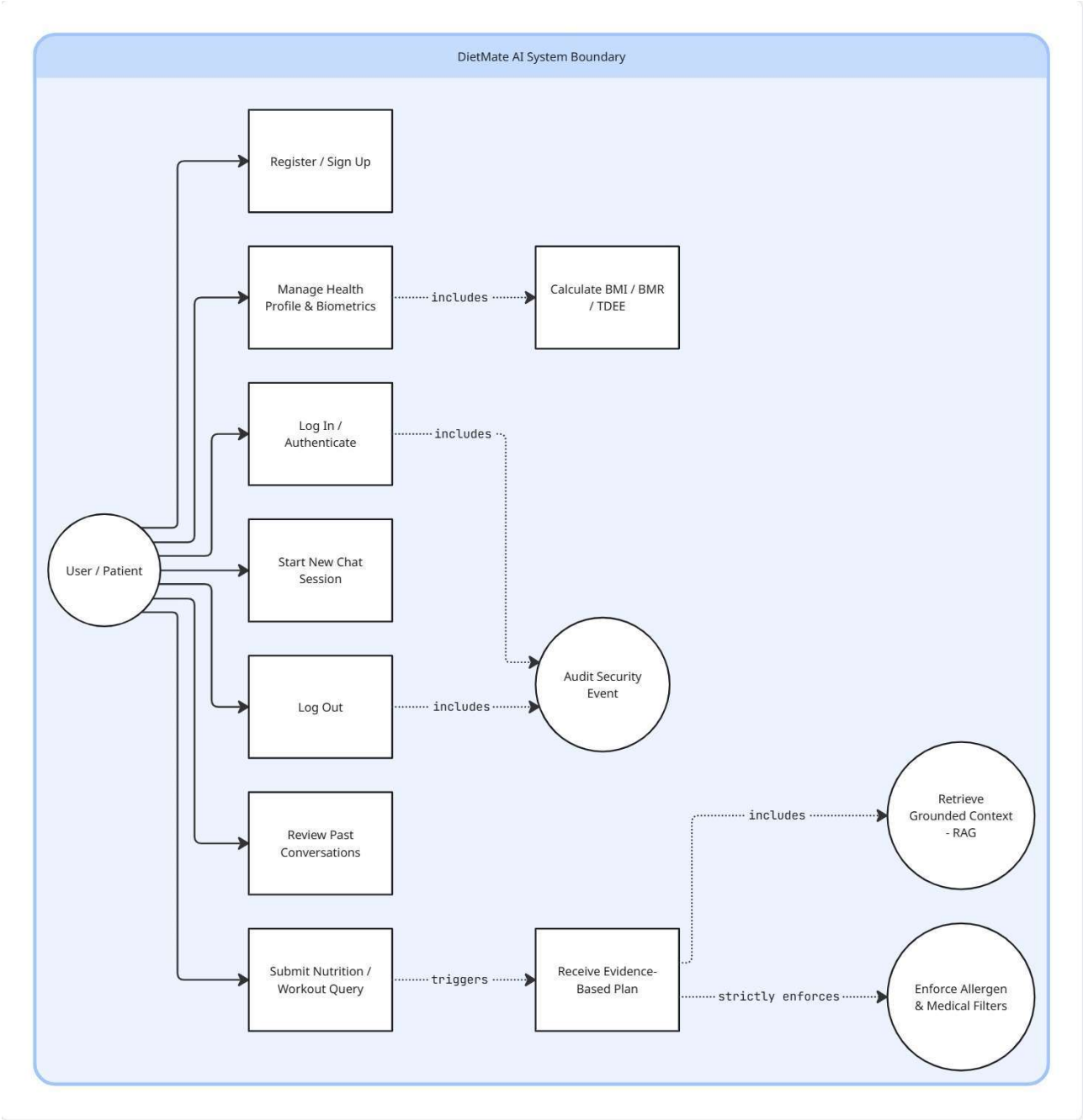
- **Relational Persistence Layer (PostgreSQL / SQLAlchemy)**: Audit trail recording login/logout events, client IP addresses, browser User-Agents, and timestamps synchronized to Indian Standard Time (IST).
- **Document Persistence Layer (MongoDB / PyMongo)**:
  - chats Collection: Stores session IDs, dynamic chat titles, creation timestamps, and last-updated timestamps.
  - messages Collection: High-throughput log storing individual message payloads, roles (user vs. bot), and IST timestamps.

Chapter 5: Conceptual Models

1. Entity Relationship Diagram (ERD):



2. Unified Modelling Language (UML) Diagram:



Chapter 6: Schema Design

1. Relational Schema (PostgreSQL)

Database name: dietmate_ai

Table Definitions:

i. user

Column Name	Data Type	Constraints
id	INTEGER	PRIMARY KEY, AUTO_INCREMENT
name	VARCHAR(100)	NOT NULL
email	VARCHAR(120)	NOT NULL, UNIQUE
password_hash	VARCHAR(255)	NOT NULL
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

ii. profile

Column Name	Data Type	Constraints
id	INTEGER	PRIMARY KEY, AUTO_INCREMENT
user_id	INTEGER	FOREIGN KEY (users.id), UNIQUE, NOT NULL
age	INTEGER	NOT NULL
gender	VARCHAR(10)	NOT NULL
height	INTEGER	NOT NULL
weight	INTEGER	NOT NULL
diet	VARCHAR(20)	NOT NULL
goal	VARCHAR(30)	NOT NULL
allergies	TEXT	NULLABLE
medical_conditions	TEXT	NULLABLE
bmi	VARCHAR(10)	NOT NULL
bmr	VARCHAR(10)	NOT NULL
target_kcal	VARCHAR(10)	NOT NULL
activity_level	VARCHAR(20)	NOT NULL
food_preferences	TEXT	NULLABLE

iii. login_history

Column Name	Data Type	Constraints
id	INTEGER	PRIMARY KEY, AUTO_INCREMENT
user_id	INTEGER	FOREIGN KEY (users.id), NOT NULL
event_type	VARCHAR(20)	NOT NULL
event_time	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP
ip_address	VARCHAR(45)	NULLABLE
user_agent	VARCHAR(255)	NULLABLE

2. Document Schema (MongoDB)

Database name: DietMate_DB

Collection definitions:

i. chats

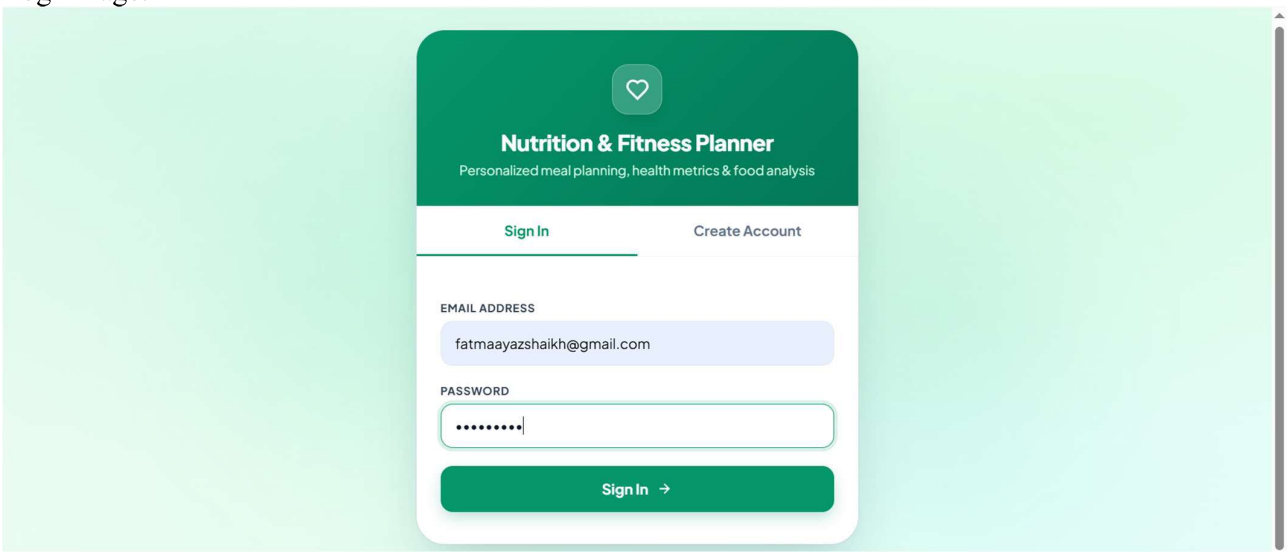
Field	BSON Type	Index
_id	ObjectId	Primary Key (Default)
chat_id	String (UUIDv4)	Unique Index
user_id	Int32	Secondary Index
title	String	None
timestamp	String / Date	None
updated_at	String / Date	Compound Index (user_id, updated_at DESC)

ii. messages

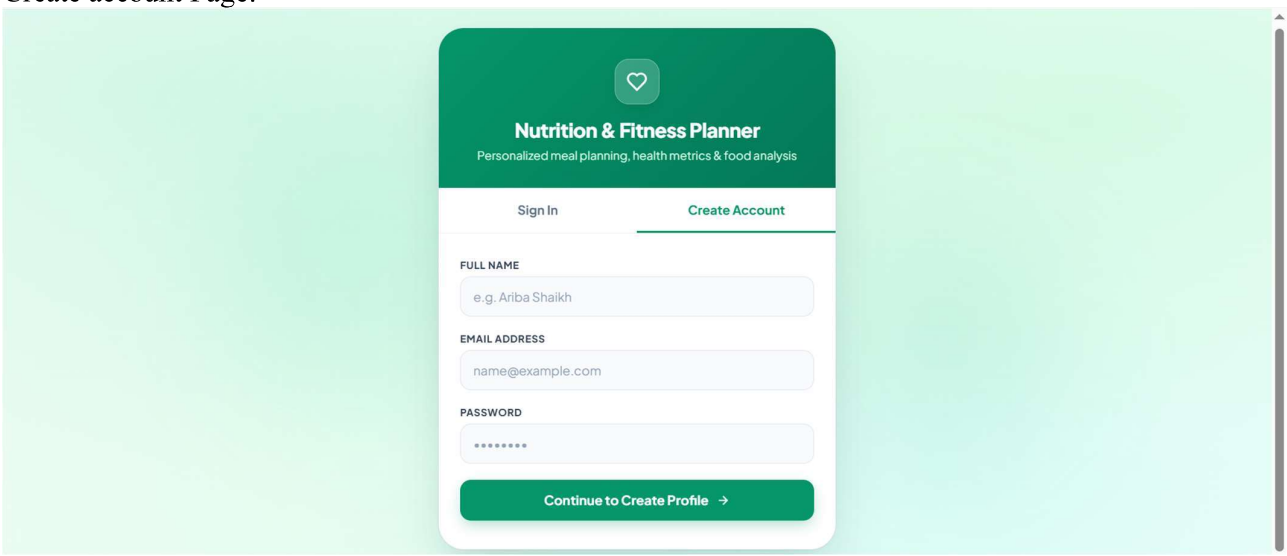
Field	BSON Type	Index
_id	ObjectId	Primary Key (Default)
chat_id	String (UUIDv4)	Index
user_id	Int32	Index
sender	String	None
message	String (Markdown)	None
timestamp	String / Date	Compound Index (chat_id, timestamp ASC)

Chapter 7: Screenshots of the User Interface

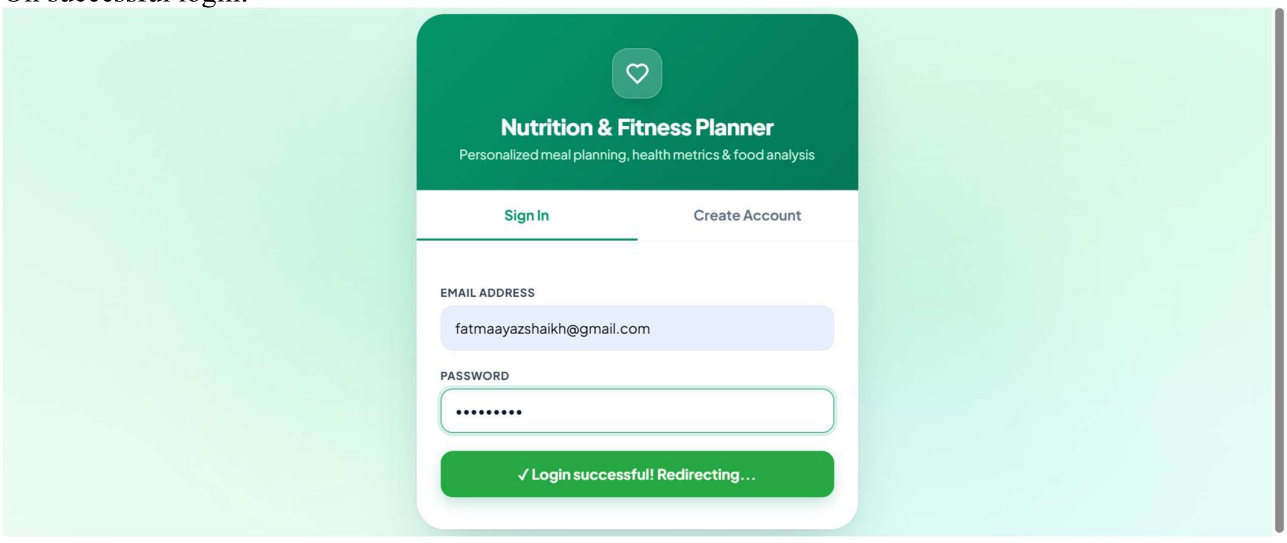
Login Page:



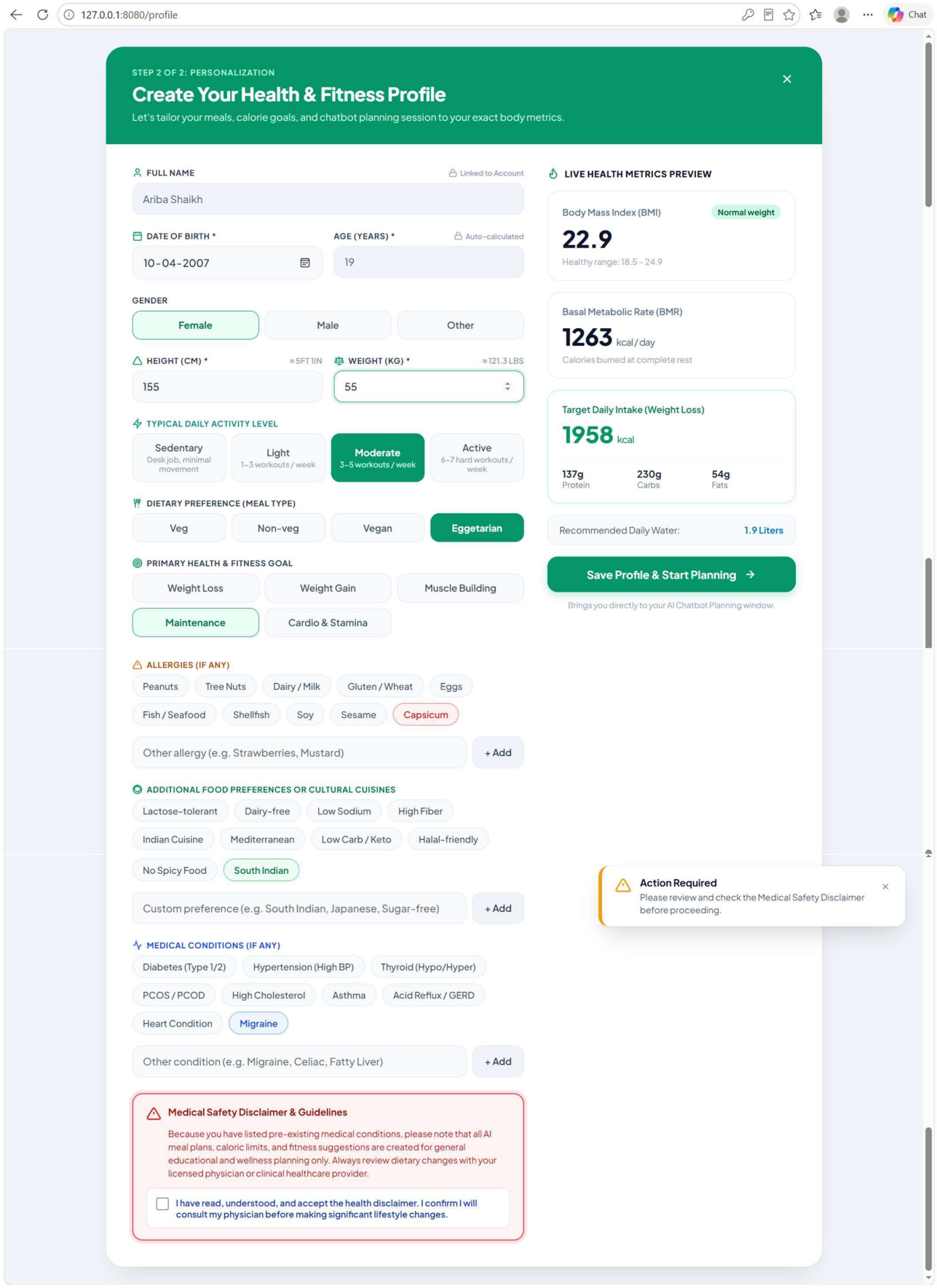
Create account Page:



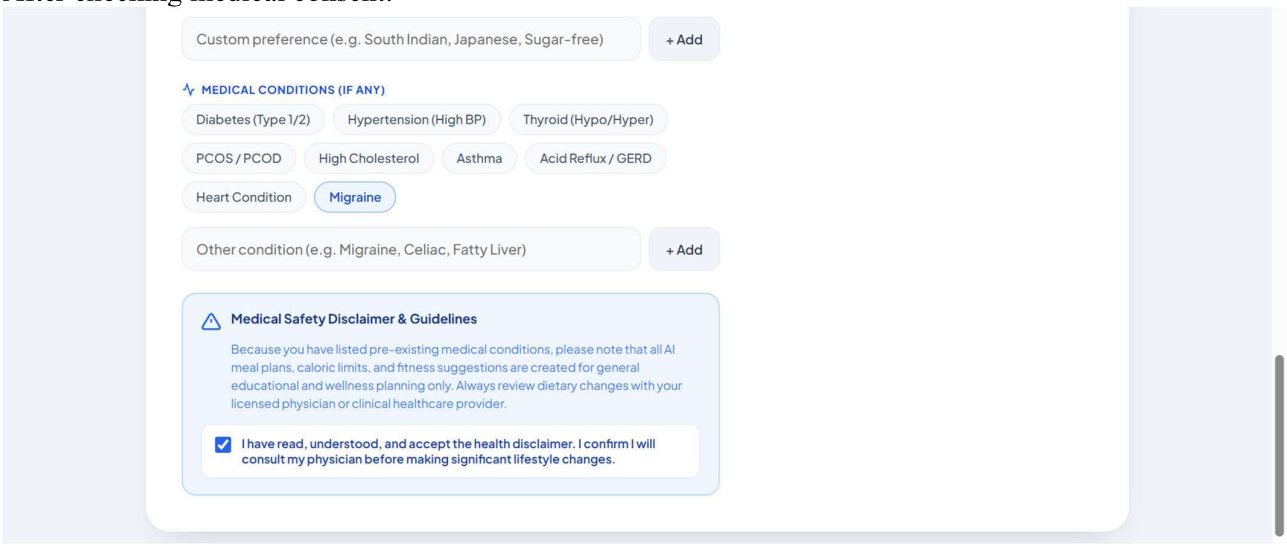
On successful login:



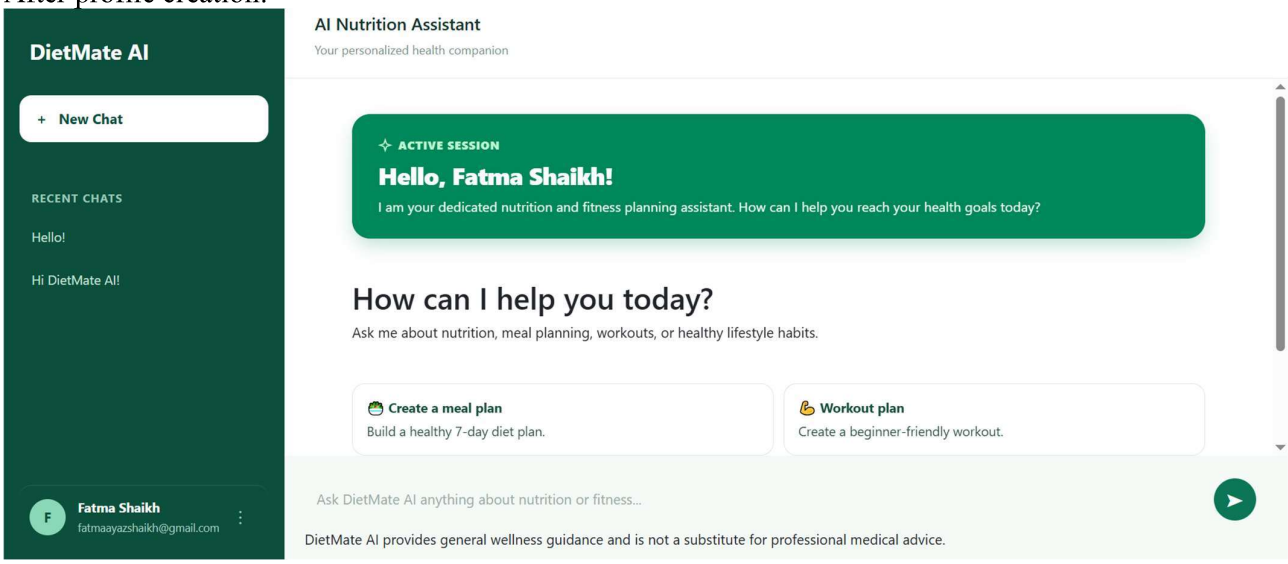
Profile creation (without checking medical consent):



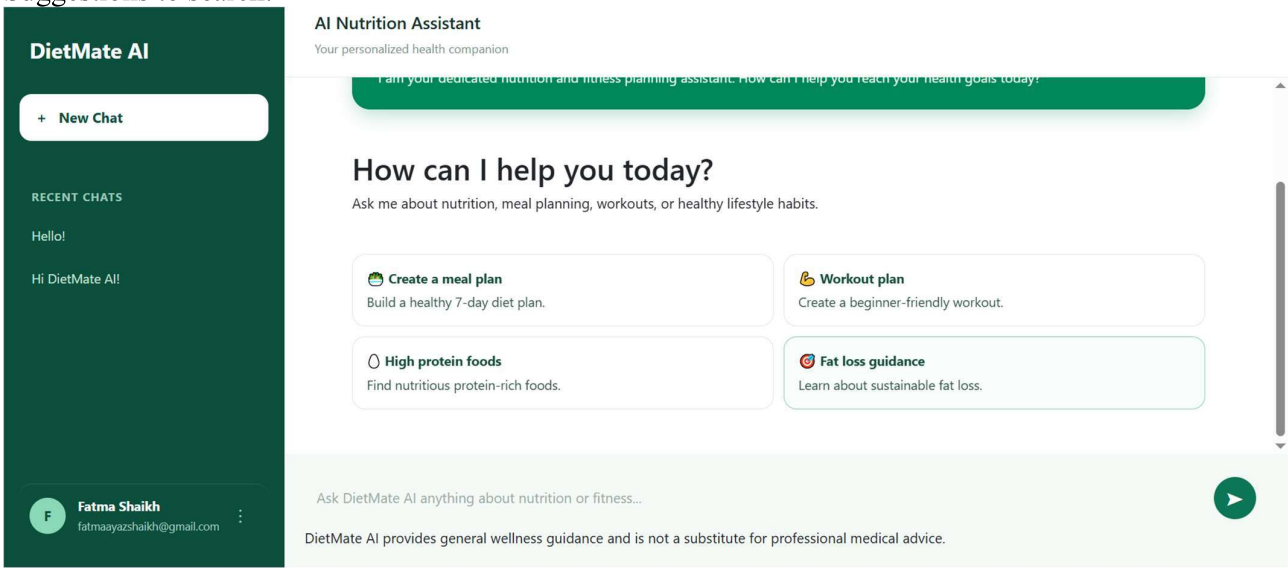
After checking medical consent:



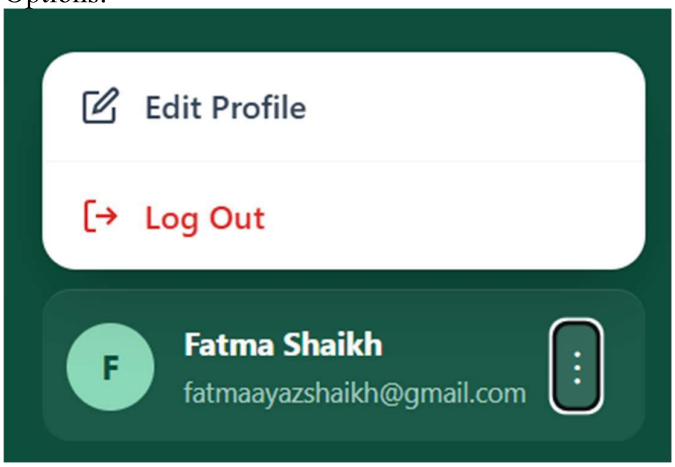
After profile creation:



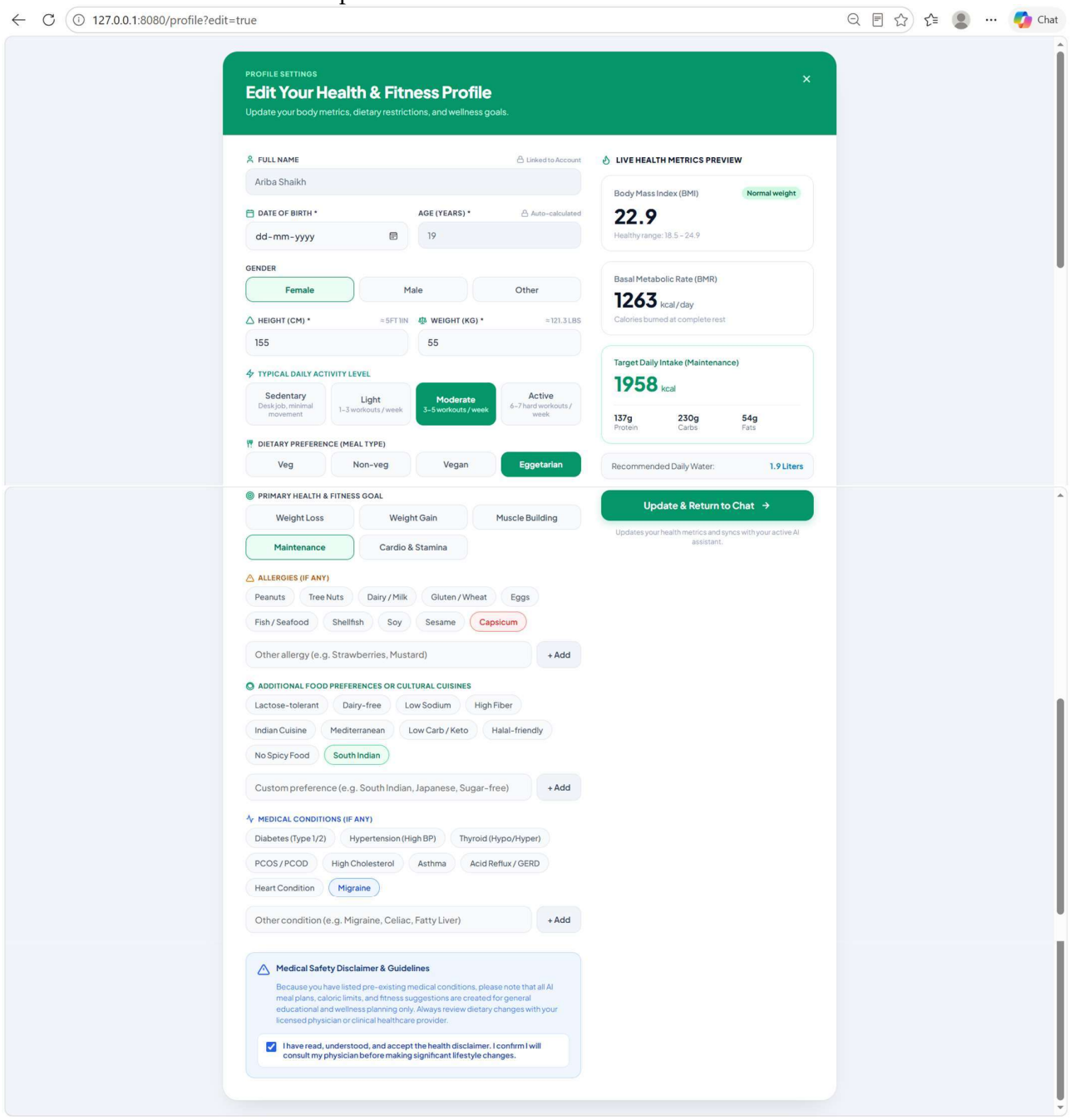
Suggestions to search:



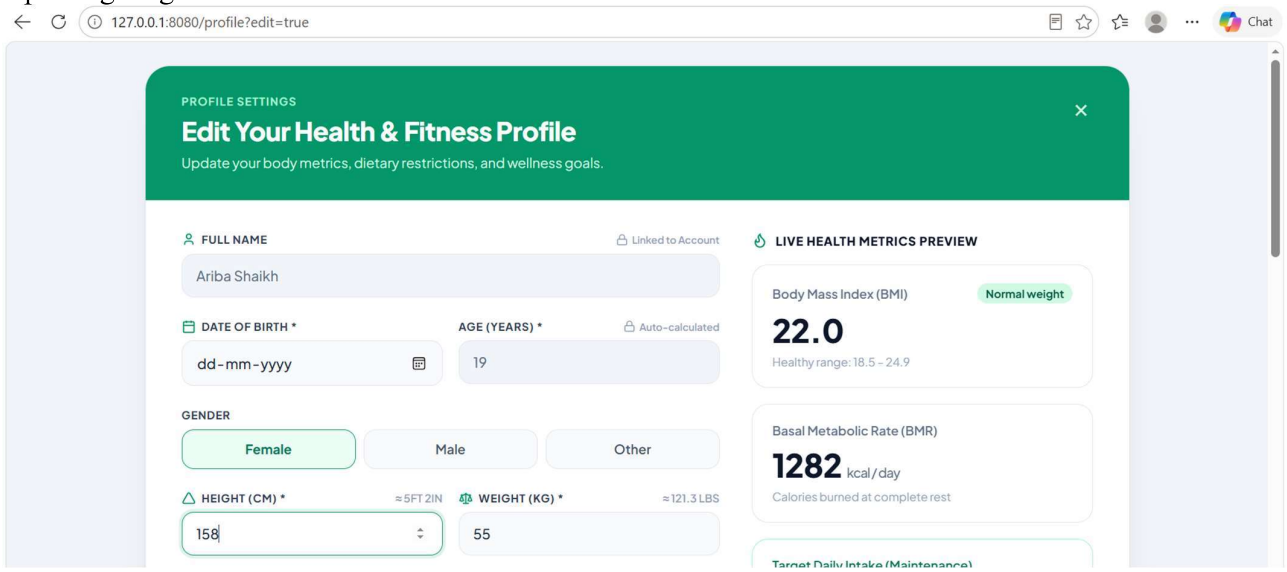
Options:



Preloaded data when clicked edit profile:



Updating height:



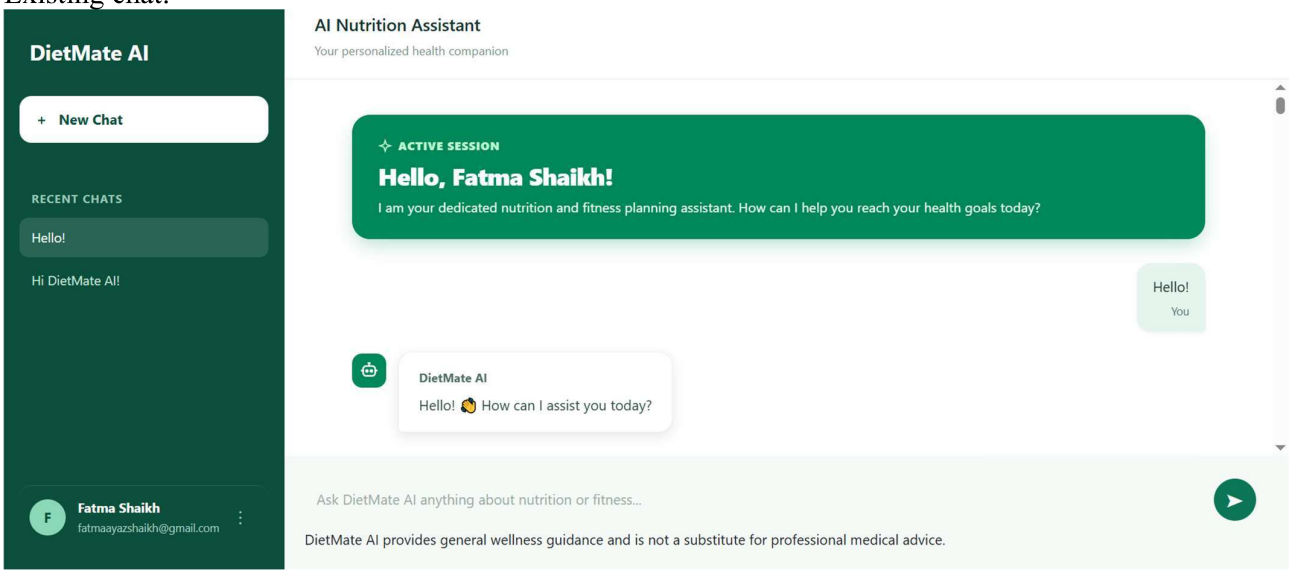
Height stored in profile table (before updating height)

	id [PK] integer	user_id integer	age integer	gender character varying (20)	height integer	weight integer	diet character varying (50)	goal character varying (50)
1	10	1	19	female	155	55	Eggetarian	Maintenance

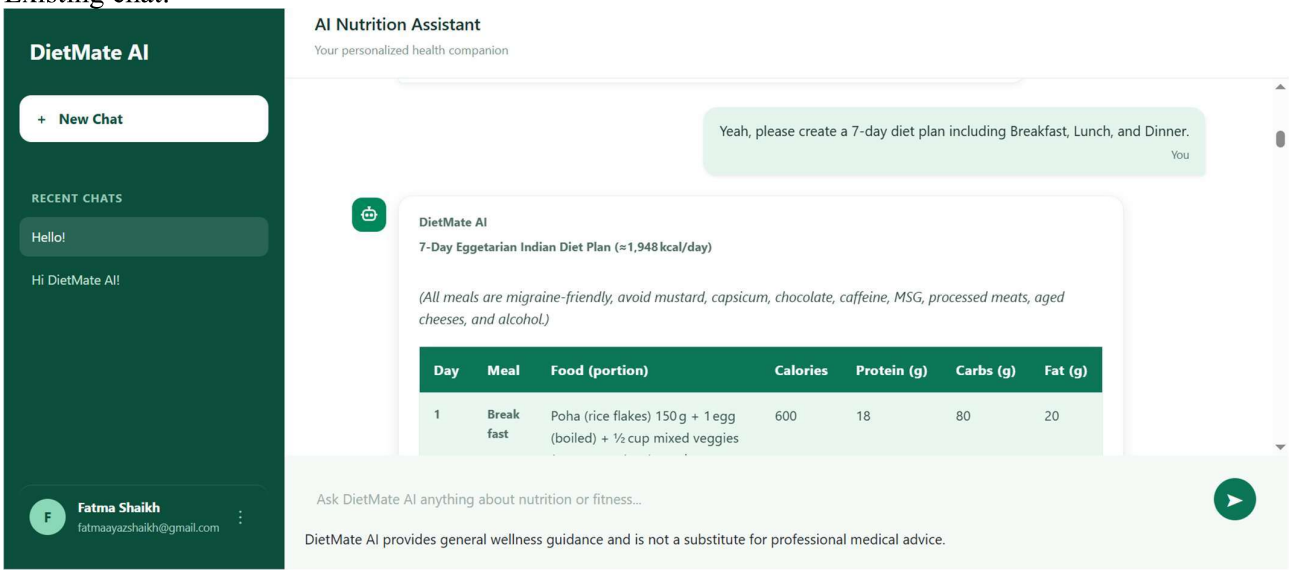
Data updated in PostgreSQL (after updating height)

	id [PK] integer	user_id integer	age integer	gender character varying (20)	height integer	weight integer	diet character varying (50)	goal character varying (50)
1	10	1	19	female	158	55	Eggetarian	Maintenance

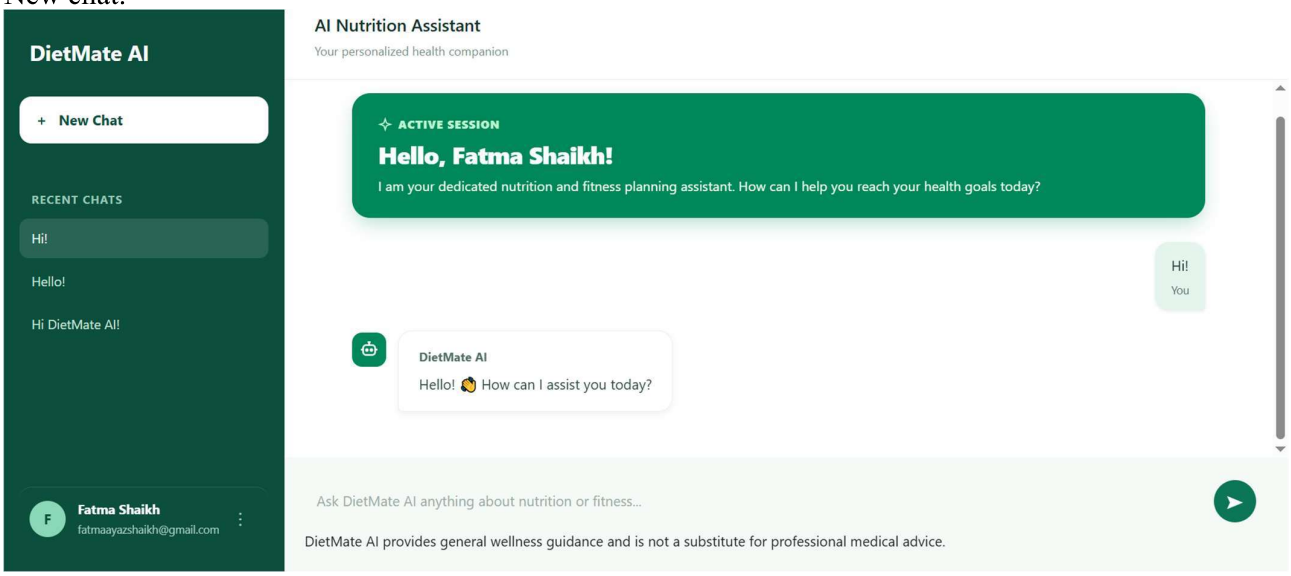
Existing chat:



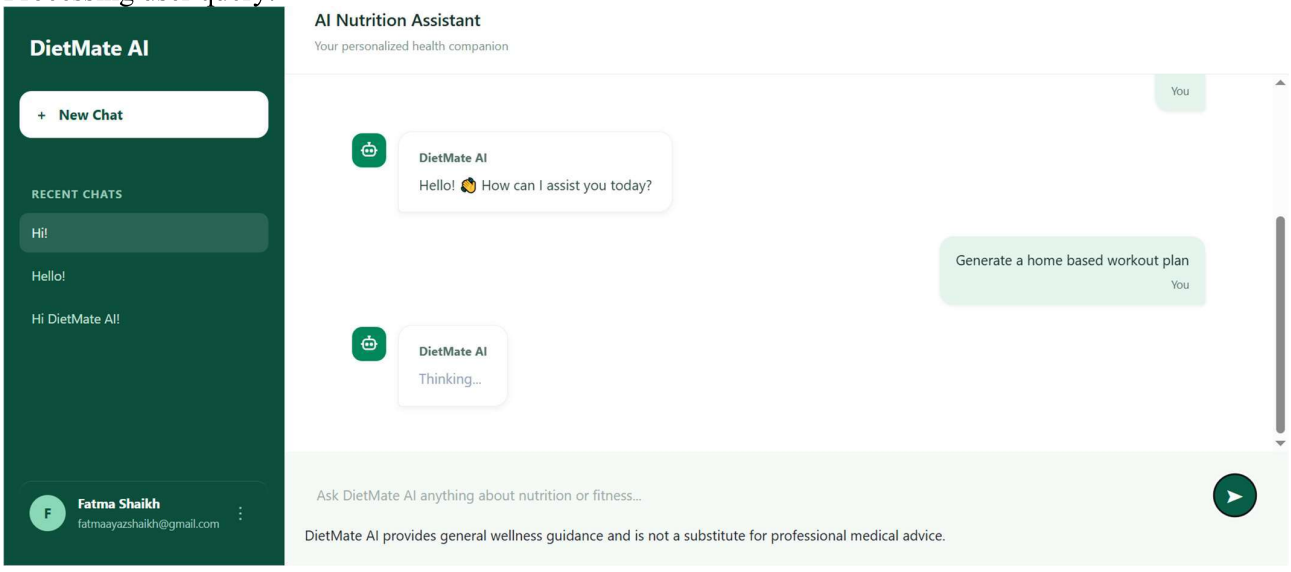
Existing chat:



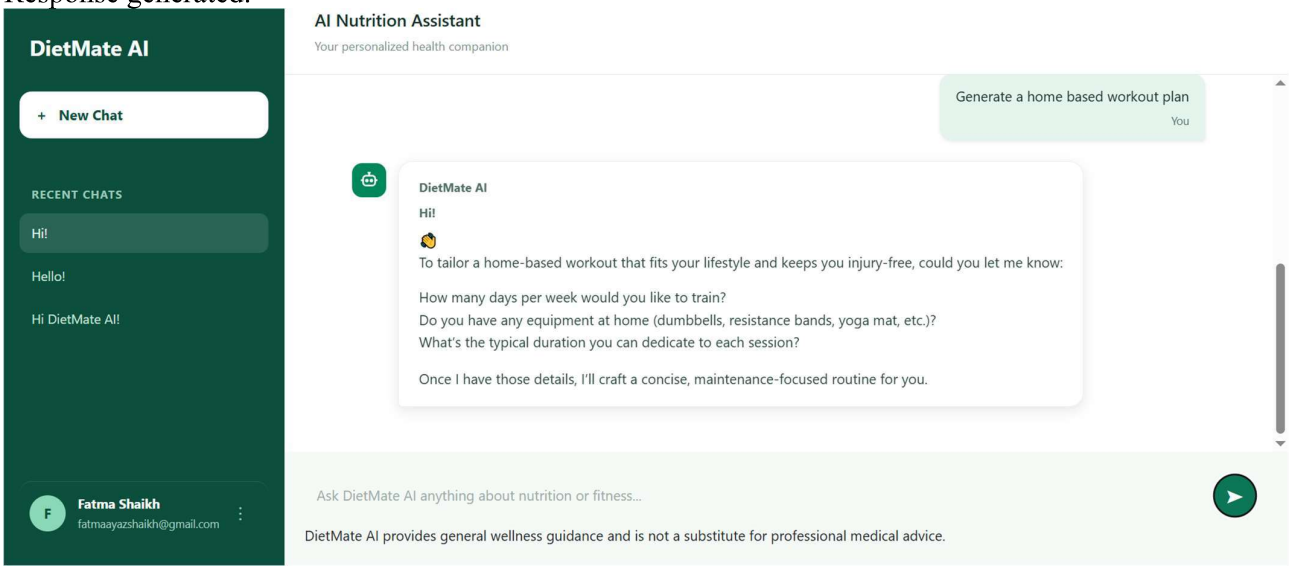
New chat:



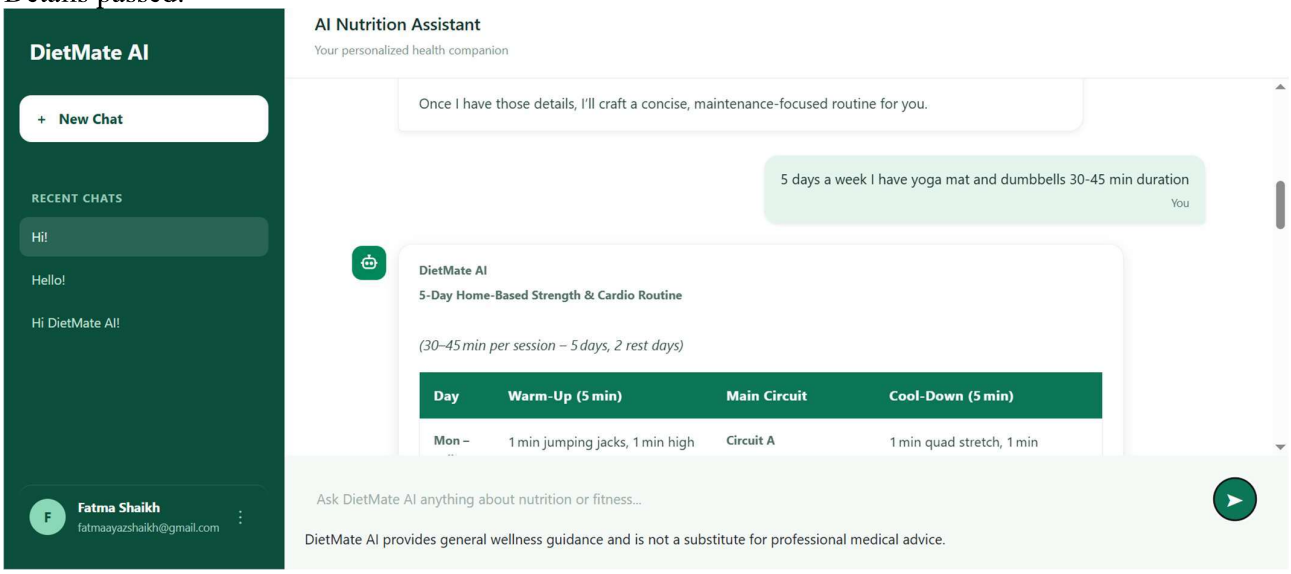
Processing user query:



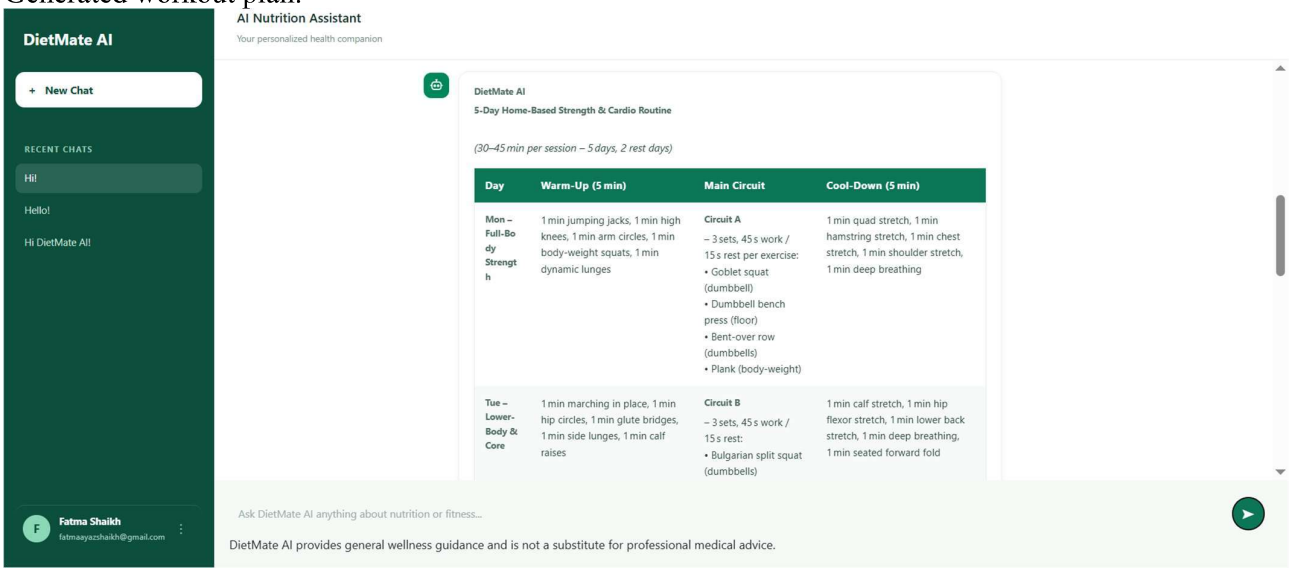
Response generated:



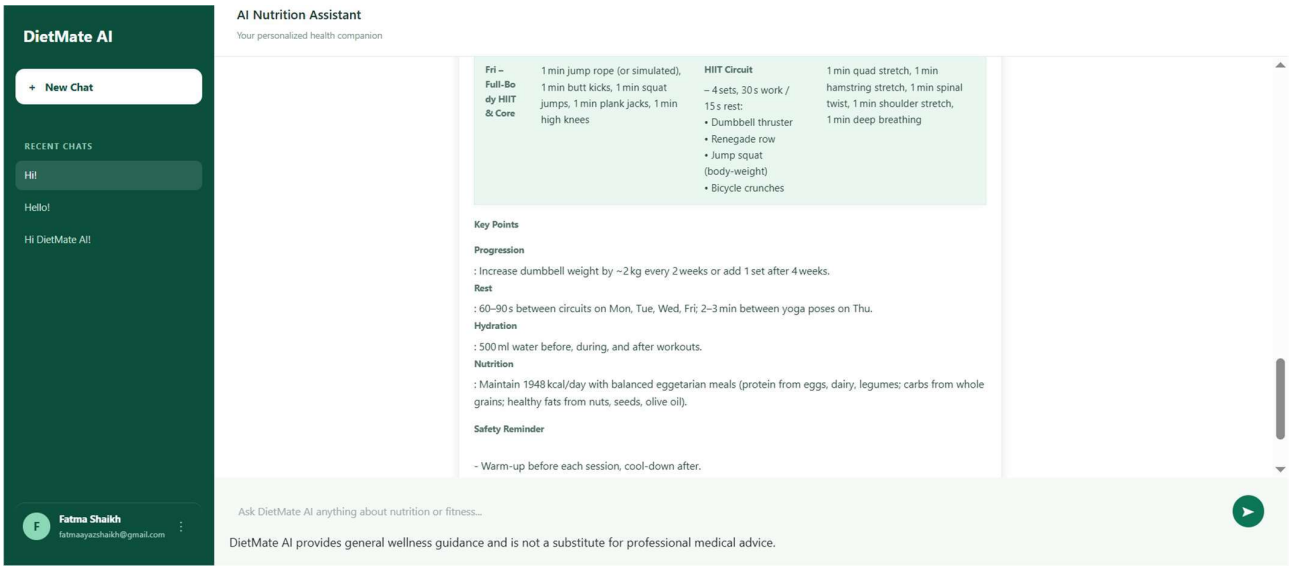
Details passed:

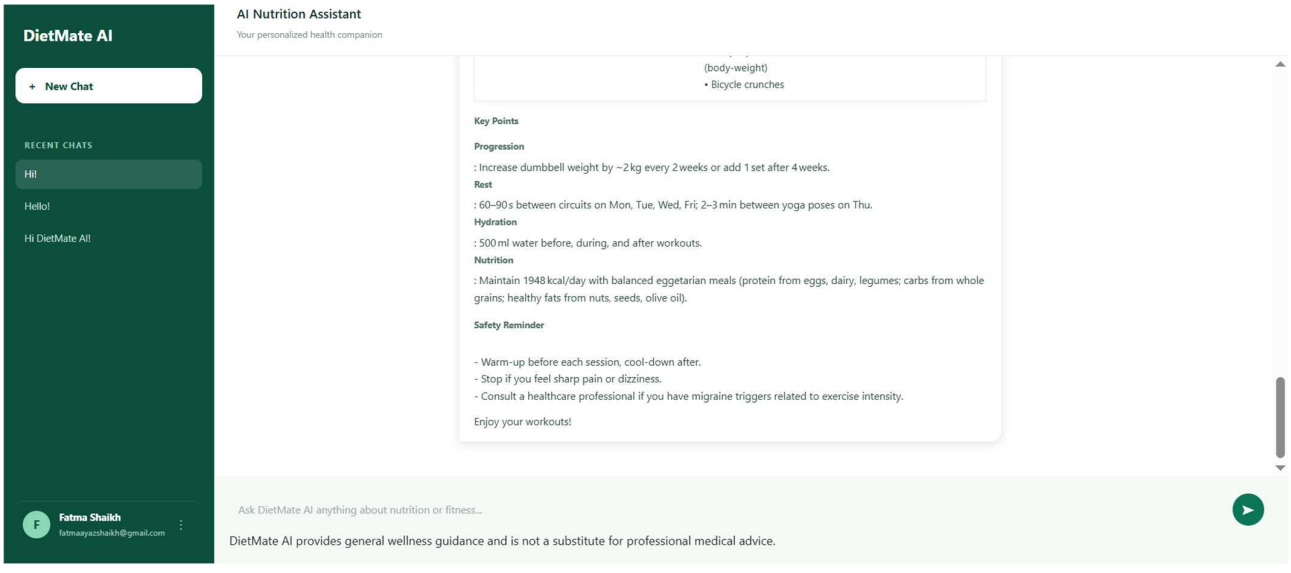


Generated workout plan:




DietMate AI
+ New Chat
RECENT CHATS
Hi!
Hello!
Hi DietMate AI!
Fatma Shaikh
fatmaayazshaikh@gmail.com
AI Nutrition Assistant
Your personalized health companion
• Side plank (30 s each side)
Wed – Upper-Body & Cardio
1 min arm swings, 1 min shoulder shrugs, 1 min push-ups (modified), 1 min mountain climbers, 1 min shadow boxing
Circuit C
– 3 sets, 45 s work / 15 s rest:
• Overhead press (dumbbells)
• Renegade row (dumbbells, push-up position)
• Triceps kickback (dumbbells)
• Burpees (no jump if needed)
1 min triceps stretch, 1 min chest opener, 1 min wrist stretch, 1 min shoulder stretch, 1 min breathing
Thu – Active Recovery / Mobility
5 min gentle walking or marching, 5 min dynamic stretches
Yoga Flow
– 20 min (hold each pose 30–45 s):
• Cat-Cow, Downward Dog, Warrior I, Warrior II, Triangle, Tree, Child's Pose, Corpse
5 min mindful breathing, gratitude reflection
Fri –
1 min jump rope (or simulated),
HIIT Circuit
1 min quad stretch, 1 min
Ask DietMate AI anything about nutrition or fitness...
DietMate AI provides general wellness guidance and is not a substitute for professional medical advice.


DietMate AI
+ New Chat
RECENT CHATS
Hi!
Hello!
Hi DietMate AI!
Fatma Shaikh
fatmaayazshaikh@gmail.com
AI Nutrition Assistant
Your personalized health companion
Fri – Full-Body HIIT & Core
1 min jump rope (or simulated), 1 min butt kicks, 1 min squat jumps, 1 min plank jacks, 1 min high knees
HIIT Circuit
– 4 sets, 30 s work / 15 s rest:
• Dumbbell thruster
• Renegade row
• Jump squat (body-weight)
• Bicycle crunches
1 min quad stretch, 1 min hamstring stretch, 1 min spinal twist, 1 min shoulder stretch, 1 min deep breathing
Key Points
Progression
: Increase dumbbell weight by ~2 kg every 2 weeks or add 1 set after 4 weeks.
Rest
: 60–90 s between circuits on Mon, Tue, Wed, Fri; 2–3 min between yoga poses on Thu.
Hydration
: 500 ml water before, during, and after workouts.
Nutrition
: Maintain 1948 kcal/day with balanced eggetarian meals (protein from eggs, dairy, legumes; carbs from whole grains; healthy fats from nuts, seeds, olive oil).
Safety Reminder
- Warm-up before each session, cool-down after.
Ask DietMate AI anything about nutrition or fitness...
DietMate AI provides general wellness guidance and is not a substitute for professional medical advice.


DietMate AI
+ New Chat
RECENT CHATS
Hi!
Hello!
Hi DietMate AI!
Fatma Shaikh
fatmaayazshaikh@gmail.com
AI Nutrition Assistant
Your personalized health companion
(body-weight)
• Bicycle crunches
Key Points
Progression
: Increase dumbbell weight by ~2 kg every 2 weeks or add 1 set after 4 weeks.
Rest
: 60–90 s between circuits on Mon, Tue, Wed, Fri; 2–3 min between yoga poses on Thu.
Hydration
: 500 ml water before, during, and after workouts.
Nutrition
: Maintain 1948 kcal/day with balanced eggetarian meals (protein from eggs, dairy, legumes; carbs from whole grains; healthy fats from nuts, seeds, olive oil).
Safety Reminder
- Warm-up before each session, cool-down after.
- Stop if you feel sharp pain or dizziness.
- Consult a healthcare professional if you have migraine triggers related to exercise intensity.
Enjoy your workouts!
Ask DietMate AI anything about nutrition or fitness...
DietMate AI provides general wellness guidance and is not a substitute for professional medical advice.

On logging out successfully:

